

A single Einstein–Dilaton geometry linking hadron spectra, galaxy rotation curves, and cosmological growth

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Abstract

A corrected, geometry-preserving formulation is given for a frozen five-dimensional Einstein–dilaton background used as a common computational input for hadronic, galactic, cosmological, and laboratory readouts. A three-stage audit first derives and seals the field equations, then compares the frozen trace with the previously declared canonical polynomial action, and finally adjudicates the result in an independent coordinate representation. The stored trace does not solve that polynomial system as labelled. It is shown, however, that the achieved warp and scalar profiles admit a healthy effective completion $S_{\text{eff}} \propto \int \sqrt{-g} (R - K(\phi)(\partial\phi)^2/2 - V(\phi))$. The reconstructed kinetic function is strictly positive on the certified radial interval, so the completion is equivalent to a canonical scalar by field redefinition. The warp and scalar profiles are preserved to 2.04×10^{-6} and 8.54×10^{-8} , respectively, while the complete field equations close at normalized floating-point residuals below 8×10^{-17} . The deformation used by the observational arms is recovered with correlation 0.999999905 and rms difference 6.88×10^{-4} . On a separately declared compact-interval probe branch, the geometry-fixed trace carrier yields the normalized matter coupling $\beta_n = \sqrt{f_g/3} f_n$. Explicitly selected Neumann endpoints give $\beta_0 = 0.0542901$ and a conditional unscreened scalar-force fraction $2\beta_0^2 = 5.89483 \times 10^{-3}$; the corresponding massless scalar-tensor branch is incompatible with Solar-System bounds and is not a viable discovery claim. The six positive modes define a correlated material-force template, but their ranges still require a physical length. No observational input enters that derivation; the interval interpretation and boundary/junction completion remain assumptions. A Liouville transformation shows that this carrier and the gauge-invariant scalar operator are the same local degree of freedom under different boundary and physical interpretations, not two towers to be added. An exhaustive NN/ND/DN/DD endpoint audit then tests the simplest boundary alternatives without observational input. NN retains the excluded massless mode; ND produces an independently verified UV-coupled ultralight mode with $m\ell = 0.00274476$ and $\beta_{\text{UV}} = 0.0542910$; and UV Dirichlet branches decouple an exact point probe. Thus changing one endpoint is not a data-independent rescue, and a physical boundary action remains necessary. The existing scalar-spectrum value $m_1/m_0 = 1.5455$ is retained as an operator-level diagnostic pending recomputation on a prospectively frozen analytic completion. The SPARC result is reported as a global in-sample calibration against its baryonic Newtonian baseline with no per-galaxy parameters, and the growth and laboratory arms as explicit dictionary-dependent projections. A corrected Ricci audit and a Wilson-scale audit further show that the legacy clock used a mislabeled radial derivative and that the stored “Wilson-loop” value is an endpoint proxy with external normalization, not a measured area law. This revision therefore preserves the operational shared-background instrument while separating its verified computational content, one conditional matter interface, and the remaining forward predictions that require additional interface actions or held-out validation. The electromagnetic audit exposes and corrects a radial-gauge error: the historical numerical kernel treated the stored domain-wall grid as a conformal coordinate. A covariant bulk-Maxwell reduction instead gives the same probability measure in both gauges, an action-derived scalar–photon vertex, and a photon Kaluza–Klein comb. Together with the scalar carrier, this forms a correlated two-sector spectral fingerprint with one still-unfixed physical length. Its central coefficients require no observational fit, but photon localization, boundary action, source, scale, and atomic sensitivity remain unselected; it is therefore a conditional prediction rather than a detection. Finally, a retrospective galaxy-level split finds that a train-only five-parameter SPARC refit improves on baryons-only Newton but is substantially worse than a train-fitted one-parameter radial-acceleration baseline. A four-bin DESI DR1 marginal growth diagnostic gives $\chi^2 = 2.692$ for the frozen dictionary and 2.419 for matched Λ CDM, providing no preference for the former.

Key words: gravitation – holography – galaxies: kinematics and dynamics – cosmology: large-scale structure of Universe – methods: numerical

1 Introduction

Einstein–scalar domain walls provide a controlled language for holographic renormalization-group flows and, with an appropriate infrared completion, for confining Yang–Mills phenomenology (DeWolfe et al., 2000; Gürsoy et al., 2008, 2011). The project studied here uses one frozen bulk profile as a shared input for a gauge-invariant scalar fluctuation operator, a global SPARC readout, a cosmological-growth dictionary, and versioned single-arm response operators. That common dependency and the accompanying machine-readable artefacts are an operational achievement: every arm can be rerun against the same frozen arrays.

Reproducibility of a common input is not, by itself, a derivation of all interfaces from one action. The stronger claim requires

two separate checks. First, the bulk arrays must solve the field equations associated with the declared action and gauge. Secondly, mappings from the holographic radius to a galaxy radius, cosmological redshift, or laboratory time must either follow from an explicit interface action or be identified as phenomenological dictionaries.

This revision performs those checks without discarding the achieved geometry. Section 2 derives the complete bulk equations. Section 3 records the three-stage consistency audit and identifies a preconditioning label mismatch. Section 4 reconstructs a positive effective kinetic function and potential that support the frozen profiles. Section 5 derives one explicitly conditional probe-matter interface without observational calibration. The remaining sections retain the established numerical readouts

with corrected evidential labels. No simultaneous multi-arm inversion or laboratory detection is claimed.

2 Effective Einstein–dilaton background

Signature $(-, +, +, +, +)$ is used together with the effective action

$$S_{\text{eff}} = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g} \left[R - \frac{1}{2} K(\phi) (\partial\phi)^2 - V(\phi) \right]. \quad (1)$$

The canonical model is the special case $K = 1$. Variation gives

$$R_{MN} = \frac{1}{2} K \partial_M \phi \partial_N \phi + \frac{1}{3} g_{MN} V, \quad (2)$$

$$K \nabla^2 \phi + \frac{1}{2} K_{,\phi} (\partial\phi)^2 = V_{,\phi}. \quad (3)$$

In domain-wall gauge,

$$ds^2 = e^{2A(u)} \eta_{\mu\nu} dx^\mu dx^\nu + du^2, \quad \phi = \phi(u), \quad (4)$$

these equations reduce to

$$A'' = -\frac{1}{6} K \phi'^2, \quad (5)$$

$$12A'^2 - \frac{1}{2} K \phi'^2 + V = 0, \quad (6)$$

$$K(\phi'' + 4A'\phi') + \frac{1}{2} K_{,\phi} \phi'^2 = V_{,\phi}. \quad (7)$$

Primes in Eqs. (5)–(7) denote d/du . The constraint alone is not a substitute for the two dynamical equations.

For comparison in conformal gauge, define $dz/du = e^{-A}$. The metric is $ds^2 = e^{2A(z)} (\eta_{\mu\nu} dx^\mu dx^\nu + dz^2)$ and the equations transform covariantly. Pure AdS has constant $A_u = \pm L^{-1}$ in domain-wall gauge but $A_z = -z^{-1}$ in the usual conformal orientation. These reference derivatives must not be mixed.

2.1 Constraint propagation and matter sources

For the canonical vacuum system, differentiating $H = 12A'^2 - \phi'^2/2 + V$ and inserting the dynamical equations gives $H' = 0$. If a density is added only through $A'' = -\phi'^2/6 - \kappa\rho$ while the vacuum constraint is retained, then

$$H' = -24\kappa A' \rho. \quad (8)$$

A consistent matter source therefore requires its action, conserved stress tensor, and corresponding constraint contribution. Equation (8) is the reason the old sourced polynomial interpretation cannot be retained unchanged.

3 Three-stage audit of the frozen trace

The trace contains 1999 samples over the stored coordinate range $[0.01, 2]$. The audit was separated into: (i) direct derivation and sealing of the equations and thresholds; (ii) comparison with hash-frozen candidate outputs; and (iii) independent integration and coordinate-gauge adjudication. Input hashes and thresholds are included in the public audit artefacts.

An independent positive control is provided by

$$W(\phi) = \frac{6}{L} + \frac{\phi^2}{2L}, \quad V = \frac{1}{2} W_{,\phi}^2 - \frac{1}{3} W^2, \quad (9)$$

Table 1: Equation-level audit of the frozen trace. Values are normalized residuals; all listed tests fail their sealed criterion.

Metric	Literal	BF sign	Criterion
ϕ kinematic rms	0.3139	0.3139	≤ 0.01
A kinematic rms	0.4615	0.4615	≤ 0.01
Constraint p95	0.03852	0.03851	≤ 0.001
Scalar EOM rms	0.3220	0.3543	≤ 0.01
Sourced warp EOM rms	0.8903	0.8903	≤ 0.01

which gives

$$V = -\frac{12}{L^2} - \frac{3\phi^2}{2L^2} - \frac{\phi^4}{12L^2} \quad (10)$$

and the exact flow

$$\phi = \phi_0 e^{(u-u_0)/L}, \quad A = A_0 - \frac{u-u_0}{L} - \frac{\phi^2 - \phi_0^2}{24}. \quad (11)$$

It has $m^2 L^2 = -3$, above the AdS_5 Breitenlohner–Freedman bound (Breitenlohner & Freedman, 1982). Independent second-order integration agrees with the exact solution to 4.92×10^{-14} ; the largest conformal-gauge residual is 3.55×10^{-15} .

Table 1 reports the same tests on the frozen trace. Both the literal stored polynomial sign and the intended BF-mass sign fail the sealed criteria. The previously reported maximum raw constraint residual, 1.81, is therefore not treated as a harmless diagnostic.

The cause is identifiable. The original numerical preconditioner rescales the state, evaluates an untransformed right-hand side, and then reverses the scaling. Consequently the stored columns satisfy approximately

$$A_{,u} \simeq u(dA)_{\text{stored}}, \quad \phi_{,u} \simeq (d\phi)_{\text{stored}}/u, \quad (12)$$

rather than being direct derivative columns. This is a provenance and label error, not evidence that the smooth warp geometry itself must be discarded.

The Bessel-based holonomic proxy in the previous version remains useful as a compact fit, but it imposes only Eq. (6). Its normalized constraint residual is 1.48×10^{-16} whereas its scalar and warp residuals are 0.9546 and 0.9494. That construction is therefore moved to the status of a descriptive proxy and is not used as the bulk action.

4 Geometry-preserving effective completion

For monotonic $\phi(u)$ and $A''(u) < 0$, Eqs. (5) and (6) can be inverted:

$$K(\phi(u)) = -\frac{6A''(u)}{\phi'(u)^2}, \quad V(\phi(u)) = -3A''(u) - 12A'(u)^2. \quad (13)$$

The scalar equation then follows from the derivative of the constraint wherever $\phi' \neq 0$. This is a geometry-preserving inverse completion: it asks which action supports the achieved profiles instead of replacing those profiles with the audit control solution.

Quintic smoothing splines remove sub-micro-scale integration noise. Ten samples at each endpoint are excluded because high-order derivatives are boundary sensitive. The certified interval is

$$0.010233 \leq u \leq 1.980981. \quad (14)$$

Table 2: Certificate for the effective completion.

Quantity	Value
Non-canonical warp max residual	2.22×10^{-16}
Constraint max residual	7.11×10^{-15}
Non-canonical scalar normalized rms	7.58×10^{-17}
Canonical warp max residual	4.44×10^{-16}
Canonical scalar max residual	1.07×10^{-14}

On this interval, the fit changes A by at most 2.04×10^{-6} and ϕ by at most 8.54×10^{-8} . The kinetic function obeys

$$8.55 \leq K(\phi) \leq 9.39 \times 10^6, \quad (15)$$

and is strictly positive at every certified sample. The wide range signals a poor normalization of the stored scalar near the ultraviolet endpoint, but not a kinetic ghost.

Defining a canonical scalar by

$$\chi'(u) = \sqrt{-6A''(u)} \quad (16)$$

gives an equivalent canonical representation. Table 2 shows closure in both field variables.

The operational deformation is also recovered. Equation (12) turns the stored expression $-u(dA)_{\text{stored}} - 1$ into the domain-wall quantity

$$\delta_{\text{eff}}(u) = -A'(u) - 1 \quad (L = 1). \quad (17)$$

Its correlation with the stored deformation is 0.999999905, with rms difference 6.88×10^{-4} . The shared observational input is therefore preserved to high precision.

5 Conditional compact-interval matter interface

The corrected bulk fixes a healthy local scalar carrier, but a holographic radial coordinate is not automatically an extra physical dimension. A blind forward benchmark is obtained by adding six explicit assumptions: one physical copy of the certified interval; a consistent Gibbons–Hawking–York/background/junction completion with no additional localized quadratic trace term; selected Neumann endpoints; probe Standard Model matter localized on a fixed slice; no brane bending or backreaction; and induced-metric coupling only at tree level. These assumptions are not selected by the frozen trace.

Let $u_{\text{phys}} = \ell u$ and define

$$\epsilon = -\frac{A''}{A'^2} > 0, \quad p = e^{4A}\epsilon, \quad w = e^{2A}\epsilon, \quad (18)$$

where the ratio defining ϵ is invariant under the constant rescaling from u to u_{phys} . The local transverse-trace action, with the normalization mapped from Arutyunov et al. (2000), is

$$S_h^{(2)} = -\frac{3\ell}{128\kappa_5^2} \int d^4x du \left[w \eta^{\mu\nu} \partial_\mu h \partial_\nu h + \frac{p}{\ell^2} (h')^2 \right]. \quad (19)$$

That calculation supplies the local quadratic normalization; it does not supply the compactification or the endpoint conditions used here.

With $p f'_n|_{\text{UV,IR}} = 0$, the dimensionless modes obey

$$-(p f'_n)' = \mu_n^2 w f_n, \quad \int du w f_n f_m = \delta_{nm}, \quad (20)$$

$$h(x, u) = \sum_n q_n(x) f_n(u), \quad \varphi_n = \frac{\sqrt{3\ell}}{8\kappa_5} q_n. \quad (21)$$

The physical trace-carrier masses are $M_n^{(h)} = \mu_n/\ell$. This tower is not an additional scalar tower independent of the ζ operator. To see this, introduce the conformal coordinate $dz_c = e^{-A} du$ and $s = e^{3A}\epsilon$. Equation (20) becomes

$$-(s f_{n,z_c})_{,z_c} = \mu_n^2 s f_n. \quad (22)$$

Writing $\Psi_n = e^B f_n$ with

$$B = \frac{1}{2} \log s = \frac{3}{2} A + \frac{1}{2} \log \epsilon = \frac{3}{2} A + \log |X| + \text{const}, \quad X = \frac{\chi'}{3A'}, \quad (23)$$

gives

$$[-\partial_{z_c}^2 + B'^2 + B''] \Psi_n = \mu_n^2 \Psi_n. \quad (24)$$

This is the local gauge-invariant scalar operator. The present Neumann compactification, the stored Dirichlet–Neumann calculation, and their use of canonical χ versus stored ϕ do not impose the same physical boundary problem, so their numerical spectra need not agree. They nevertheless describe one scalar degree of freedom and cannot be double counted. In the holographic reading u is an RG coordinate and the poles are composite; in the compact reading it is a physical interval and the modes are four-dimensional scalars.

For a conserved probe stress tensor, the longitudinal trace-sector field decouples and the transverse trace contributes

$$\delta S_m = \int d^4x \sqrt{-g} \frac{hT}{8} = \sum_n \int d^4x \sqrt{-g} \frac{\beta_n}{M_{\text{Pl}}} \varphi_n T. \quad (25)$$

Tensor reduction on the same interval gives

$$M_{\text{Pl}}^2 = \frac{\ell I_g}{\kappa_5^2}, \quad I_g = \int du e^{2A}, \quad (26)$$

and therefore

$$\beta_n(u_m) = \sqrt{\frac{I_g}{3}} f_n(u_m). \quad (27)$$

The unknown κ_5 and the interval length cancel from the dimensionless coupling, although ℓ remains necessary to fix every nonzero force range.

The blind numerical reduction gives $I_g = 0.8079601$ and $I_w = 91.37501$. The Neumann zero mode is $f_0 = I_w^{-1/2}$, so

$$\beta_0 = \sqrt{\frac{I_g}{3I_w}} = 0.0542901, \quad 2\beta_0^2 = 5.89483 \times 10^{-3}. \quad (28)$$

If this massless branch survives the boundary completion, two unscreened nonrelativistic bodies have

$$V_{12}(r) = -\frac{G_N m_1 m_2}{r} (1 + 2\beta_0^2). \quad (29)$$

Thus the scalar term is 0.589 per cent of the tensor Newtonian term in this benchmark. Under the standard universal massless scalar-tensor mapping it would give

$$\gamma = \frac{1 - 2\beta_0^2}{1 + 2\beta_0^2} = 0.988279, \quad (30)$$

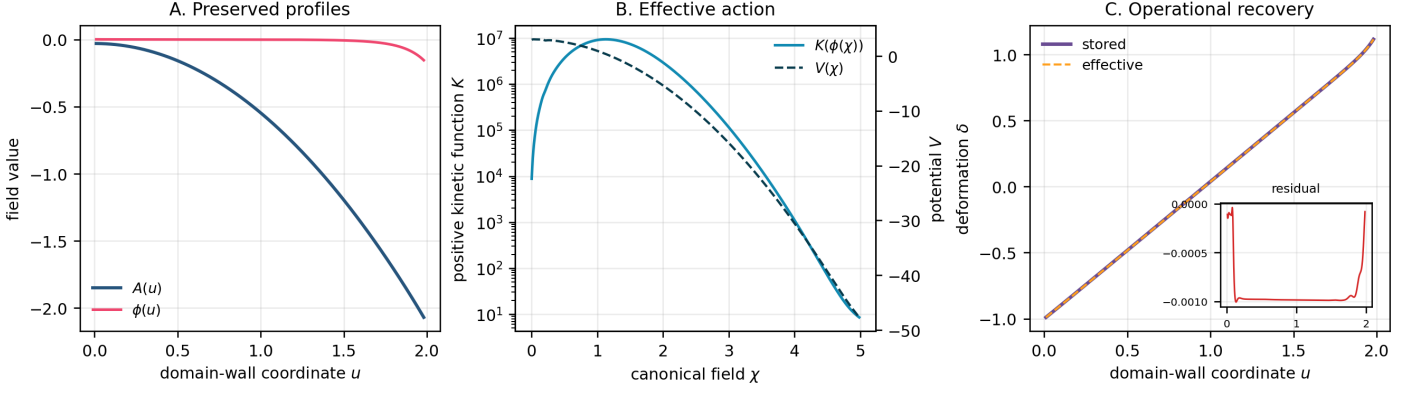


Figure 1: Geometry-preserving effective completion. Left: reconstructed warp and scalar profiles on the certified interval. Centre: positive kinetic function and potential in canonical field coordinates. Right: stored and reconstructed operational deformations and their residual.

whereas Cassini found $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti et al., 2003). The massless unscreened Neumann branch is therefore excluded. A different boundary sector may lift or remove the zero mode, but it must be specified prospectively and its complete spectrum recomputed.

The computed dimensionless tower is

$$(\mu_0, \dots, \mu_3) = (0, 0.913899, 1.541695, 2.133778), \quad (31)$$

$$(\mu_4, \dots, \mu_6) = (2.728899, 3.336654, 3.954473), \quad (32)$$

with UV-slice couplings

$$(\beta_0, \beta_1, \beta_2) = (0.0542901, 0.000705106, 0.00155651), \quad (33)$$

$$(\beta_3, \beta_4) = (0.00233763, 0.00282007), \quad (34)$$

$$(\beta_5, \beta_6) = (0.00306861, 0.00320280). \quad (35)$$

The finite-element result passes mesh, orthonormality, residual, and zero-mode checks; a separate shooting solve provides the independent numerical check.

For the six positive modes of the current Neumann basis, $\alpha_n = 2\beta_n^2$ sums to 7.20230×10^{-5} and the correlated mass ratios are

$$1 : 1.686942 : 2.334807 : 2.985996 : 3.651009 : 4.327036. \quad (36)$$

For $x = r/\ell$ they define the material-force templates

$$\frac{a_\varphi}{a_N} = \sum_{n>0} \alpha_n (1 + \mu_n x) e^{-\mu_n x}, \quad (37)$$

$$\frac{|\partial_r a_\varphi|}{|\partial_r a_N|} = \sum_{n>0} \alpha_n \left(1 + \mu_n x + \frac{\mu_n^2 x^2}{2} \right) e^{-\mu_n x}. \quad (38)$$

A mechanical mode with angular frequency ω_m and quality factor Q then responds to a source-modulated differential acceleration as

$$|\Delta x(\Omega)| = \frac{|\Delta a(\Omega)|}{\sqrt{(\omega_m^2 - \Omega^2)^2 + (\Omega \omega_m / Q)^2}}. \quad (39)$$

Equations (38)–(39) are a transfer law, not a predicted signal: a source trajectory, ℓ , detector geometry, elastic response, noise, and mode occupation must still be supplied.

For a Maxwell field localized on a four-dimensional slice, $T_{\text{EM}} = 0$ and classical Weyl invariance gives no independent local $\varphi_n F_{\mu\nu} F^{\mu\nu}$ vertex from the conformal induced metric. This statement does not apply to a photon propagating through the fifth dimension: its action also samples the perturbed radial lapse,

whose constraint-generated coupling is derived below. A universal rescaling of all masses does not by itself change leading clock-frequency ratios, and anomaly or non-universal coefficients remain separate possible channels (Damour & Donoghue, 2010).

6 Data and evaluation protocol

Table 3 records the numerical cross-check against the original PDF. The values are retained; only their evidential interpretation is tightened. In particular, the original SPARC analysis was not benchmarked against state-of-the-art dark-matter or modified-gravity fits. Its declared comparator was the baryonic Newtonian baseline under the same evaluation protocol, and that comparison scope is not broadened here.

6.1 SPARC rotation curves

SPARC provides 175 galaxies with $3.6 \mu\text{m}$ photometry, gas contributions, and resolved rotation curves (Lelli et al., 2016). The final model curve does not read the observed velocity v_{obs} during forward evaluation. Its five global fitted parameters were, however, selected by differential evolution on the same 175 curves. The present result is therefore described as a global *in-sample calibration with no per-galaxy parameters*, not as a held-out prediction. The frozen global vector is

$$\mathcal{A} = 0.13983, \quad n = 2.21605, \quad m = 1.20433, \quad (40)$$

$$\gamma = 0.23356, \quad \Sigma_0 = 0.60488. \quad (41)$$

The original ranking loss is retained for exact reproducibility,

$$\mathcal{L}_{\text{rank}} = \sum_i \frac{[v_{\text{model}}(r_i) - v_{\text{obs}}(r_i)]^2}{v_{\text{obs}}(r_i)^2 + (1 \text{ km s}^{-1})^2}, \quad (42)$$

but is no longer denoted χ^2 because its denominator is not an observational variance. The conventional uncertainty-weighted statistic is reported separately.

6.2 Growth and BOSS DR12

The cosmological arm uses the explicit dictionary

$$z_{\text{trace}} = 1 + z_{\text{cos}}, \quad N = -\ln z_{\text{trace}}. \quad (43)$$

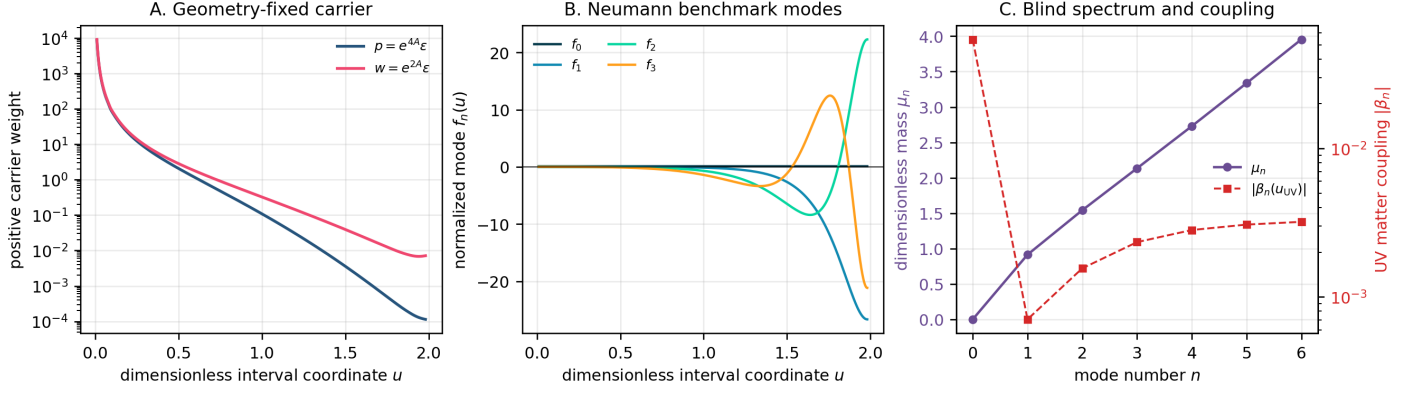


Figure 2: Blind conditional matter completion. Left: positive carrier weights fixed by the reconstructed geometry. Centre: normalized Neumann benchmark modes. Right: dimensionless trace-carrier masses and UV-slice matter couplings. No observational residual or historical fitted coupling enters this calculation.

Table 3: Cross-check of the principal numerical results against the original PDF. ED/Newton denotes the ED-anchored and baryonic Newtonian readouts; dict/ Λ CDM denotes the dictionary model and its matched flat- Λ CDM reference.

Quantity	Original PDF	Revision	Evidential status
Stored scalar ratio	1.5455	1.5454665	operator diagnostic
IR scale proxy	0.203 GeV^2 ; 451 MeV; 1.60 GeV	unchanged	external normalization
SPARC ranking medians (ED/Newton)	1.86/1.93	1.862/1.928	in-sample
SPARC uncertainty medians (ED/Newton)	831/906	831/906	in-sample
SPARC wins	150/175; 149/175	unchanged	same two metrics
BOSS covariance χ^2 (dict/ Λ CDM)	2.266/2.443	unchanged	statistically indistinguishable
High- z growth shift	≈ -12 per cent	unchanged	composite-dictionary prediction

It is not derived from a compactification or a four-dimensional FRW action. Predictions that use Eq. (43) are consequently labelled dictionary-dependent. The comparison uses the BOSS DR12 covariance at $z = \{0.38, 0.51, 0.61\}$ (Alam et al., 2017) and a matched Planck-2018 flat- Λ CDM reference with $\Omega_{m0} = 0.315$, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and $\sigma_{8,0} = 0.811$ (Planck Collaboration, 2020).

7 Gauge-invariant scalar operator

For a self-adjoint fluctuation problem,

$$-(p\psi')' + q\psi = \lambda w\psi, \quad (44)$$

normalizable solutions with declared ultraviolet and infrared boundary conditions define a discrete finite-domain spectrum. The stored scalar channel uses a Schrödinger potential $V_\zeta = B'^2 + B''$, with $B = 3A/2 + \log X$, Dirichlet ultraviolet and Neumann infrared conditions.

The operator calculation gives

$$(m_1/m_0)_{\text{stored}} = 1.5454665 \quad (45)$$

and agrees between its two covariantly transformed numerical realizations at the 10^{-7} level. This establishes internal coordinate consistency of the stored operator calculation. A physical glueball claim additionally requires recomputation using the canonical field χ , a confining infrared completion, normalizability, and prospectively fixed boundary conditions (Gürsoy et al., 2008, 2011; Morningstar & Peardon, 1999). The stored value is compatible with the broad early estimate 1.54(11) of Morningstar & Peardon (1999), but the modern continuum SU(3) values quoted by Athenodorou & Teper (2020) imply $m_{0^{++}}/m_{0^{++}} =$

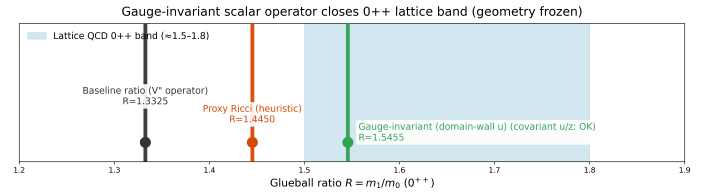


Figure 3: Stored scalar 0⁺⁺ operator ratio on the frozen profile. In this revision $m_1/m_0 = 1.5455$ is retained as a reproducible operator diagnostic; it is not yet promoted to a physical glueball prediction of the reconstructed effective action.

1.7195(160). The stored ratio is about 10.1 per cent lower and is not a sub-percent lattice prediction.

7.1 Infrared scale proxy

The stored quantity

$$\sigma_{\text{proxy}} = \frac{e^{2A(u_{\text{IR}})}}{2\pi\alpha'} \quad (46)$$

uses the finite endpoint rather than a demonstrated world-sheet turning point. With $e^{2A(u_{\text{IR}})} = 1.471 \times 10^{-2}$ and the externally chosen $\alpha' = 1.1527 \times 10^{-2} \text{ GeV}^{-2}$, it gives $\sigma_{\text{proxy}} = 0.203 \text{ GeV}^2$ and $\sqrt{\sigma_{\text{proxy}}} = 451 \text{ MeV}$. Applying the external scalar-channel factor $c = 3.55$ gives $m_0 \simeq 1.60 \text{ GeV}$. These are model-dependent scale conversions, not parameter-free predictions. The actual causal direction is therefore frozen ED endpoint plus external α' and c , followed by σ_{proxy} and m_0 ; no Wilson loop feeds the geometry. The linked lattice-gauge artefact currently supplies elementary plaquettes and a plaquette correlator,

not rectangular $W(R, T)$, a static potential, or a Creutz area-law extraction. Moreover a true holographic loop requires the string frame. Identifying the reconstructed χ with a standard IHQCD dilaton would give $A_s = A \pm \chi/\sqrt{6}$; neither orientation has a smooth interior minimum on the certified interval. An endpoint tension therefore additionally assumes a hard wall. Even a future lattice result $a^2\sigma$ fixes ℓ only if an independent matching relation ℓ/a is derived.

8 Global SPARC readout

Let $x = r/r_{\max}$ and let $\Sigma_{b,\text{norm}}$ be the normalized baryonic surface-brightness proxy. The retained readout is

$$\delta_{\text{tot}} = \text{clip}[(1-w)\delta_{\text{eff}}(u(x)) + w\delta_{\text{miss}}, 0, 0.4], \quad (47)$$

$$w = x^\gamma, \quad (48)$$

$$\delta_{\text{miss}} = \mathcal{A} \left(1 + \frac{\Sigma_{b,\text{norm}}}{\Sigma_0} \right)^{-n} x^m, \quad (49)$$

$$v_{\text{model}} = v_{\text{bar}} \sqrt{1 + \delta_{\text{tot}}}. \quad (50)$$

All galaxies share the same global parameter vector. Under the in-sample evaluation, the median ranking loss is 1.862 for the ED-anchored readout and 1.928 for the Newtonian comparator; the corresponding uncertainty-weighted medians are 831 and 906. The ED readout wins in 150/175 galaxies under the ranking loss and 149/175 under the uncertainty-weighted statistic.

These numbers are reproducible performance measurements of the declared readout. A decisive predictive test requires freezing the complete pipeline and evaluating galaxies not used for global parameter selection.

9 Dictionary-dependent cosmological growth

Under Eq. (43), a scalar-derived equation-of-state template is inserted into a matter-plus-dark-energy background. The standard linear growth equation is

$$D_{,NN} + \left(2 + \frac{d \ln H}{d \ln a} \right) D_{,N} - \frac{3}{2} \Omega_m(N) D = 0, \quad (51)$$

with $f\sigma_8 = \sigma_{8,0}(d \ln D/d \ln a)D$. At the three BOSS DR12 redshifts, the stored covariance calculation gives

$$\chi_{\text{dict}}^2 = 2.266, \quad \chi_{\Lambda\text{CDM}}^2 = 2.443. \quad (52)$$

Thus $\Delta\chi^2 = -0.177$, a statistically negligible difference at current precision. The stored high-redshift suppression, approximately -12 per cent near $z_{\text{cos}} = 1$, is a falsifiable prediction of the *composite dictionary model*, not of the five-dimensional bulk action alone.

10 Prediction-factory diagnostics

The revised calculation is organized as a fail-closed graph of physical links: an output is promoted only when the action, boundary data, dimensional scale, and evaluation protocol required by the corresponding arrow are present. Four new checks illustrate the distinction between generating a prediction and retrofitting a result.

Boundary branches. All four separated self-adjoint endpoint choices were solved with the same bulk coefficients and without observational input. The NN branch has its exact zero mode. Replacing only the IR condition gives an ND lightest mode

$$\mu_{\text{ND},0} = 0.00274498, \quad \beta_{\text{UV},0} = 0.05429095. \quad (53)$$

An independent continuous-shooting calculation gives $\mu_{\text{ND},0} = 0.0027447613$, a relative difference 7.84×10^{-5} . The former zero mode is therefore ultralight rather than removed. DN and DD start at $\mu = 0.91382$ and 1.23053 , respectively, but an exact UV point probe obeys Dirichlet data and decouples. No branch is selected: that requires a boundary or junction action fixed before testing.

The minimal positive Robin boundary-potential family makes one candidate action explicit, with no boundary kinetic terms:

$$Q[f] = \int du p(f')^2 + b_{\text{UV}} f_{\text{UV}}^2 + b_{\text{IR}} f_{\text{IR}}^2, \quad b_{\text{UV}}, b_{\text{IR}} \geq 0. \quad (54)$$

It implies $-p f'_{\text{UV}} + b_{\text{UV}} f_{\text{UV}} = 0$ and $p f'_{\text{IR}} + b_{\text{IR}} f_{\text{IR}} = 0$. The dimensionless scan coordinate is $\rho_i = b_i/C_p$, where $C_p = (\int du/p)^{-1} = 6.8836329 \times 10^{-4}$. A prospective two-parameter scan finds that IR stiffness alone cannot raise the lightest mode above $\mu_0 = 0.002744976$. UV stiffness raises that mass continuously. At much larger stiffness the first pair undergoes an avoided crossing and exchanges UV residue near $\rho_{\text{UV}} \simeq 1.12 \times 10^5$; the minimum sampled first-pair gap is 0.0119229 . The independently finite-differenced identity

$$\frac{\partial \mu_n^2}{\partial b_{\text{UV}}} = f_n(u_{\text{UV}})^2 = \frac{3\beta_n^2}{I_g} \quad (55)$$

is the Hellmann–Feynman link between boundary stiffness, pole motion, and material residue. The map makes the consequences of any prospectively selected boundary coefficients falsifiable; it does not select them.

Retrospective SPARC split. A deterministic galaxy-level split assigns 122 galaxies to training, 26 to validation, and 27 to test. The present trace-backed P5 formula was refitted on training galaxies only; validation and test were then read once. On the test group (621 velocity points), its diagonal χ^2 per point is 203.26 versus 254.30 for baryons-only Newton, and P5 wins in 22/27 galaxies. A galaxy-cluster bootstrap nevertheless gives a 95 per cent interval $[-15.26, 81.65]$ for the P5-minus-Newton log-likelihood improvement per point, which crosses zero. A one-global-parameter radial-acceleration relation (McGaugh et al., 2016), fitted on the same training split, obtains $\chi^2/\text{point} = 60.99$; P5 wins only 8/27 and its relative log-likelihood per point is -71.13 with interval $[-199.42, -7.72]$. Moreover, the P5 optimizer exhausted its fixed budget and four parameters reached their search bounds. All three models use the same historical unsigned baryonic quadrature, so their comparison is aligned, but the very large absolute χ^2 values and previously exposed catalogue make this a retrospective development diagnostic rather than confirmation.

Published DESI DR1 entries. Four published ShapeFit-plus-BAO marginal entries inside the frozen trace domain, at $z_{\text{eff}} = \{0.30, 0.51, 0.71, 0.92\}$, were evaluated without refitting (DESI Collaboration et al., 2025). Comparing their quoted $f\sigma_8$

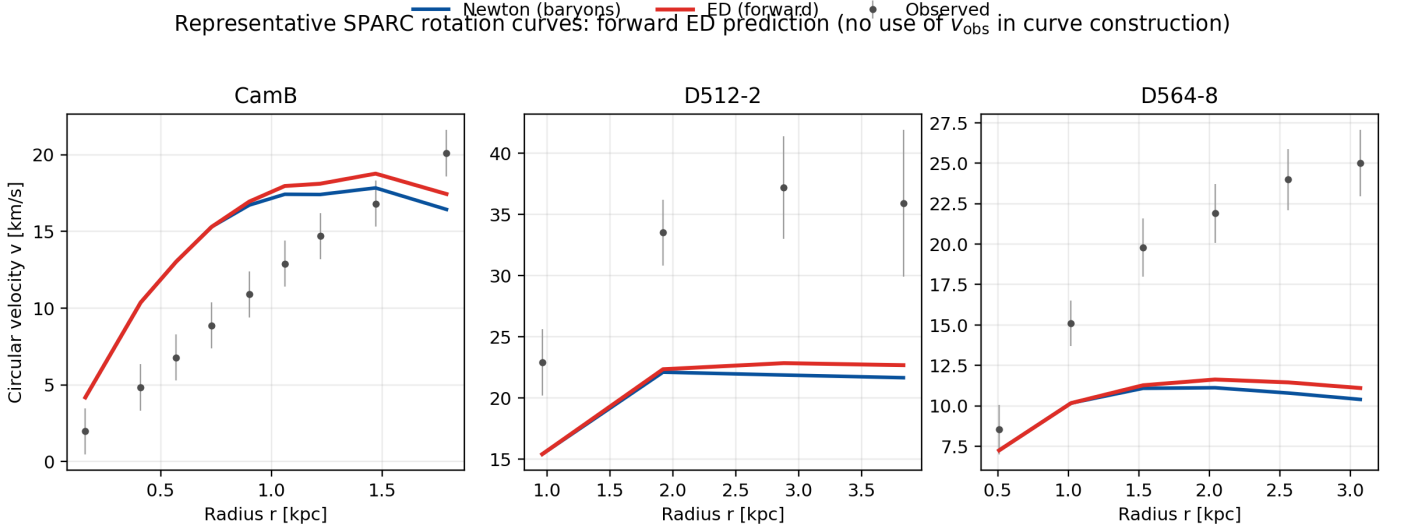


Figure 4: Representative SPARC curves. The model curve is forward-evaluated from baryonic inputs after the global readout parameters have been calibrated on the full sample; observed velocities are overlaid for evaluation.

marginals directly with the frozen $f\sigma_8$ curves gives

$$\begin{aligned} \chi^2_{\text{dict,diag}} &= 2.6917, & \chi^2_{\Lambda\text{CDM,diag}} &= 2.4189, \\ \Delta\chi^2 &= +0.2729. \end{aligned} \quad (56)$$

Both curves are compatible with this low-dimensional diagnostic and the small positive difference does not favour the dictionary curve. Because distances, shape-slope information, and within-vector and cross-bin correlations are not modeled, these numbers are explicitly not the official DESI likelihood.

Wilson route. The new analyser validates $SU(3)$ link matrices and implements rectangular $W(R, T)$, effective potentials, Creutz ratios, blocking, and jackknife uncertainties. It fails closed on the current inputs: the available lattice files contain plaquette/correlator summaries, not link configurations. Consequently neither $a^2\sigma$ nor a continuum string tension is reported.

11 Single-arm response-operator spectroscopy

Let $\{\psi_n\}$ denote the stored response-operator basis and let δa be a vector of modal amplitudes. Each observational arm α has a versioned response operator

$$J_{in}^{(\alpha)} = \frac{\partial y_i^{(\alpha)}}{\partial a_n}, \quad \delta y^{(\alpha)} \simeq J^{(\alpha)} \delta a. \quad (57)$$

The DESI example reconstructs a bounded δG from one residual vector. This is an operational single-arm inverse problem, not a simultaneous multi-arm detection. This stored basis is not identified with the independently normalized trace-carrier modes f_n of Section 5.

For diagnostic comparison, normalized arm blocks are stacked and decomposed as

$$\hat{J}_{\text{multi}} = U \Sigma V^T, \quad (58)$$

after Frobenius-per-output scaling. The singular values summarize shared sensitivity directions; they are not a global reconstruction.

A modeled time dependence may be written

$$\dot{a}_n = \sum_m M_{nm} a_m + S_n, \quad (59)$$

where both M and the closure source S must be recorded. Observable projections such as $\Delta f = \sum_n \kappa_n a_n$ remain conditional on their declared coupling ansatz.

12 Dictionary-based geometric clock and legacy laboratory interface

The five-dimensional domain-wall curvature is

$$R_{5,\text{phys}} = \frac{\hat{R}_5}{\ell^2}, \quad \hat{R}_5 = -8A'' - 20A'^2 = \frac{1}{2}\chi'^2 + \frac{5}{3}V. \quad (60)$$

A fresh audit verifies the second, independent Einstein-trace identity to 2.84×10^{-14} and reproduces the same scalar pointwise after transforming to conformal radial coordinates. It also finds that the stored clock evaluated curvature by treating the solver field $(dA)_{\text{stored}}$ as A' , although the reproduced trace satisfies $A' = u(dA)_{\text{stored}}$. On their common certified slice the stored curvature spans -21.60 to -19.09 , while the corrected value spans -13.25 to -73.47 ; their rms difference is 26.91. The old plot below is retained solely as provenance.

A corrected geometry-only protocol factor can instead be declared as

$$g_R(u) = \frac{e^{A(u)} \sqrt{|\hat{R}_5(u)|}}{e^{A_0} \sqrt{|\hat{R}_{5,0}|}}, \quad \Theta_R(u) = \int_1^u g_R(s) ds. \quad (61)$$

Equation (61) is a deterministic dimensionless curvature phase, not proper time and not a mode amplitude. On a physical compact branch it would define $\omega_R = c\sqrt{|\hat{R}_5|}/\ell$, while $\omega_n = c\mu_n/\ell$; hence ω_n/ω_R is geometry-fixed but seconds still require ℓ . The historical LOCK5 instead inferred ξ from the two frequencies it subsequently “verified” and mixed this bulk curvature with $R_4 \simeq 8\pi G\rho/c^2$. Its location table is therefore a circular calibration, not a Ricci-clock prediction.

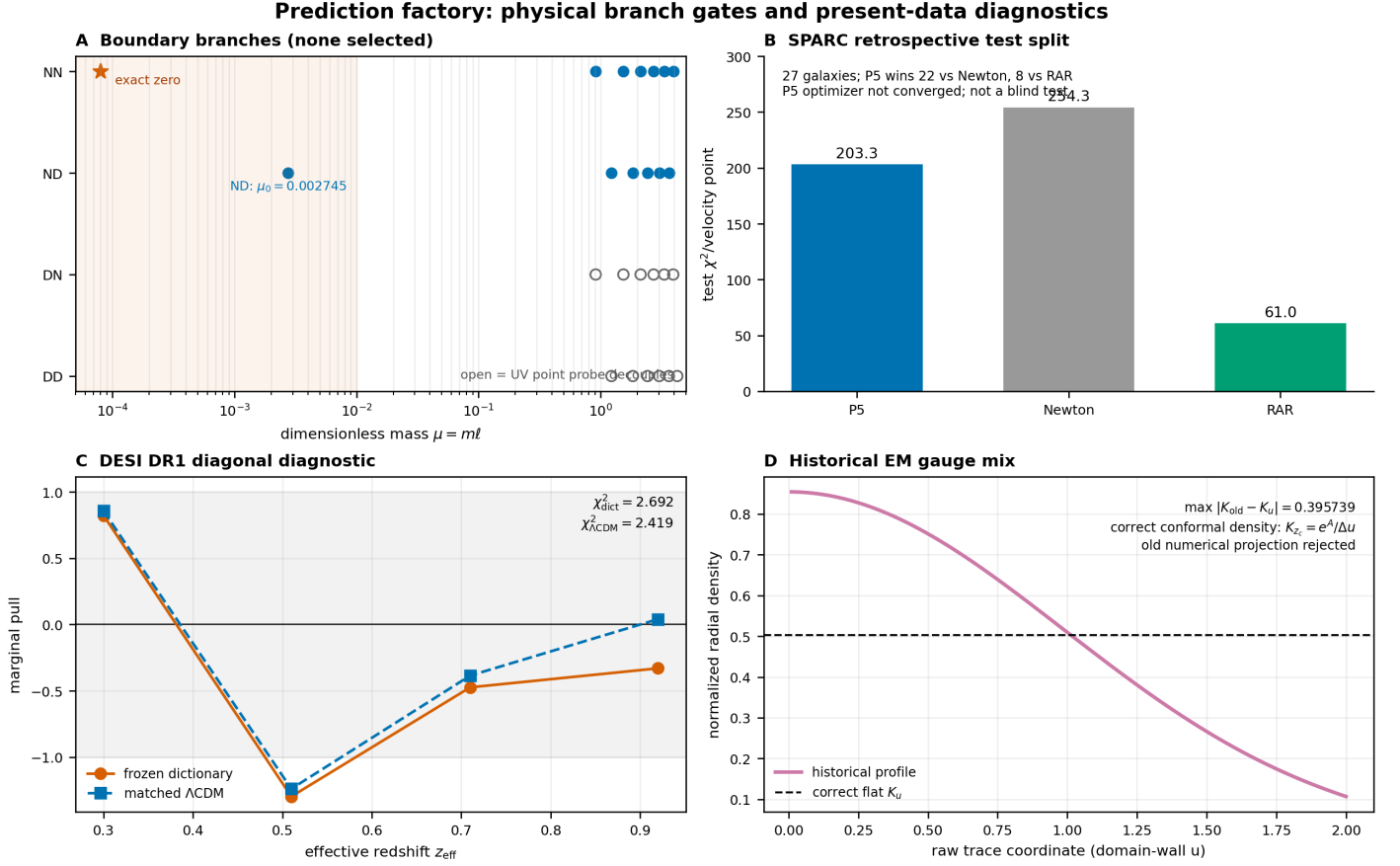


Figure 5: Prediction-factory diagnostics. (A) The four endpoint branches: filled markers couple to the declared UV point probe, while open markers obey UV Dirichlet data; no branch is selected. (B) Absolute SPARC test-split scores show that P5 improves on baryons-only Newton but underperforms the train-fitted RAR comparator. (C) Published DESI DR1 marginal pulls give no preference over matched Λ CDM. (D) The historical electromagnetic profile is compared with the coordinate-correct $Z = 1$ domain-wall zero-mode density; their disagreement exposes the old radial-gauge substitution.

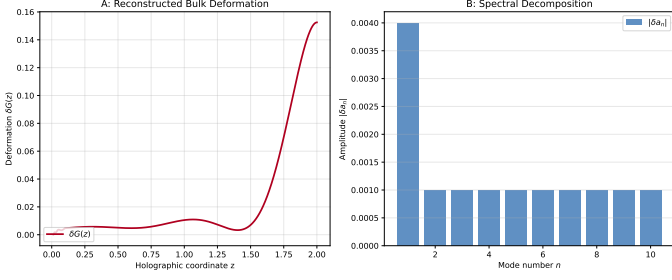


Figure 6: Bounded single-arm reconstruction and modal amplitudes. The plotted deformation is response-operator dependent.

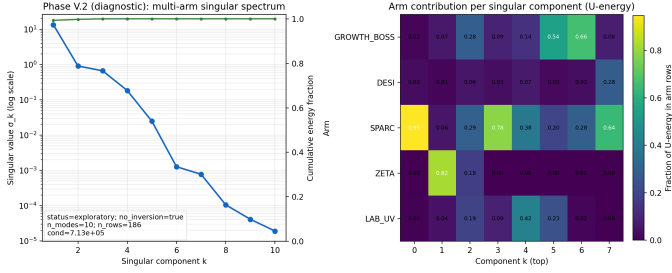


Figure 7: SVD of the stacked normalized response blocks. Diagnostic only; no cross-arm closure is inferred.

The historical UV-screened laboratory channel was tabulated on the raw trace grid, labelled that grid as conformal z , and used

$$K_{\text{old}}(z) = \frac{e^{A(z)}}{\int e^{A(s)} ds}, \quad y_{\text{old}}(t) = \int_{z_{\text{UV}}}^{z(t)} K_{\text{old}}(s) \delta(s) ds. \quad (62)$$

The effective-action audit identifies that raw grid as domain-wall u . Constructing the conformal coordinate instead requires $dz_c = e^{-A(u)} du$ and gives a conformal span 4.90458, not the raw-grid span. For a flat photon with $Z = 1$, the coordinate-correct domain-wall density is constant, $K_u = 0.507422$, whereas the historical profile differs from the corresponding full-grid constant by as much as 0.395739. Thus the previous near-machine-precision “recovery” was an identity produced by setting $A = \log K_{\text{old}}$, not a check against the reconstructed geometry.

For the stored Yb comparison ($n = 595$), the historical projection gives $\chi^2/n = 22.59$ and Pearson $r = -0.0554$. This null/poor-fit historical comparison remains useful negative provenance, but it is not a test of the coordinate-correct response and does not constitute a clock detection.

A consistent completion starts from the bulk action

$$S_\gamma = -\frac{1}{4g_5^2} \int d^5x \sqrt{-g} Z(\chi) F_{MN} F^{MN}. \quad (63)$$

For $ds^2 = e^{2A(u)} \eta_{\mu\nu} dx^\mu dx^\nu + \ell^2 du^2$, the dimensionless photon eigenvalues and profiles obey

$$-\partial_u(e^{2A} Z \partial_u f_{\gamma,n}) = \mu_{\gamma,n}^2 Z f_{\gamma,n}, \quad \int du Z f_{\gamma,m} f_{\gamma,n} = \delta_{mn}, \quad m_{\gamma,n} = \mu_{\gamma,n}/\ell. \quad (64)$$

In conformal gauge the same equations contain the weight $e^A Z$

and the two normalized densities are

$$K_u = \frac{Z |f_\gamma|^2}{\int du Z |f_\gamma|^2}, \quad K_{z_c} = \frac{e^A Z |f_\gamma|^2}{\int dz_c e^A Z |f_\gamma|^2}, \quad K_u du = K_{z_c} dz_c. \quad (65)$$

The cumulative probability measures agree to 6.27×10^{-12} . Equation (65) is the covariant functional form underlying the old ansatz, but Eq. (62) was not its numerical realization.

The scalar-photon vertex is also fixed at this conditional action level. In the same almost-radial gauge used for the trace carrier, the linear constraint of Arutyunov et al. (2000) gives

$$A_u h_{uu} = \frac{1}{4} \partial_u h, \quad N = \frac{1}{2} h_{uu} = \frac{\partial_u h}{8A_u}. \quad (66)$$

The four-dimensional conformal trace cancels from $\sqrt{-g} F_{\mu\nu} F^{\mu\nu}$, but the radial lapse N does not. For $Z = e^{c_\gamma \chi}$, comoving endpoints, no brane kinetic terms, and $h = \sum_n q_n f_n$, dimensional reduction therefore gives

$$B_F = 1 + \sum_n d_{\gamma n} \frac{\varphi_n}{M_{\text{Pl}}} + \dots, \quad d_{\gamma n}(c_\gamma) = \sqrt{\frac{I_g}{3}} \frac{\int du e^{c_\gamma \chi} f'_n / A_u}{\int du e^{c_\gamma \chi}}. \quad (67)$$

No coefficient is fitted at $c_\gamma = 0$. For the six positive NN modes, $d_{\gamma n} = (3.94563, -2.52337, 2.90124, -2.42675, 2.83547, -2.44815)$. Mode signs are conventional, while the source products $\beta_n d_{\gamma n}$ are invariant. The unfitted response to a nonconstant gauge kinetic function is separately fixed by

$$\partial_{c_\gamma} d_{\gamma n}|_0 = \sqrt{\frac{I_g}{3}} \text{Cov}_u \left(\chi, \frac{f'_n}{A_u} \right). \quad (68)$$

Neumann photon data at $c_\gamma = 0$ give the independent dimensionless comb

$$\mu_{\gamma,n} = (0, 0.652597, 1.301427, 1.939255, \dots), \quad \eta_{\gamma,n} = (1, 0.61936, 0.69848, 0.75808, \dots), \quad (69)$$

where $\eta_{\gamma,n}$ is the squared UV charge residue relative to the zero mode. With the additional interface assumption of a conserved four-dimensional electric current localized at the UV endpoint, $S_J = q_5 \int d^4x A_\mu(x, u_{\text{UV}}) J^\mu$, its minimally coupled static template is

$$V_\gamma(r) = V_{\text{Coulomb}}(r) \left[1 + \sum_{n>0} \eta_{\gamma,n} e^{-\mu_{\gamma,n} r / \ell} \right]. \quad (70)$$

Together with the scalar masses, this is a two-sector or “double-comb” fingerprint tied to the same ℓ . For a point source in the stored sign convention, the corresponding fine-structure response is

$$\frac{\delta \ln \alpha}{U} = \sum_n 2\beta_n d_{\gamma n} e^{-\mu_{\gamma,n} r / \ell}, \quad U = \frac{GM}{r}. \quad (71)$$

An atomic comparison would then require $\delta \ln(\nu_A/\nu_B) = \Delta K_\alpha \delta \ln \alpha$. Equations (69)–(71) do not predict hertz until ℓ , a source, atomic sensitivities, and the physical bulk-photon and

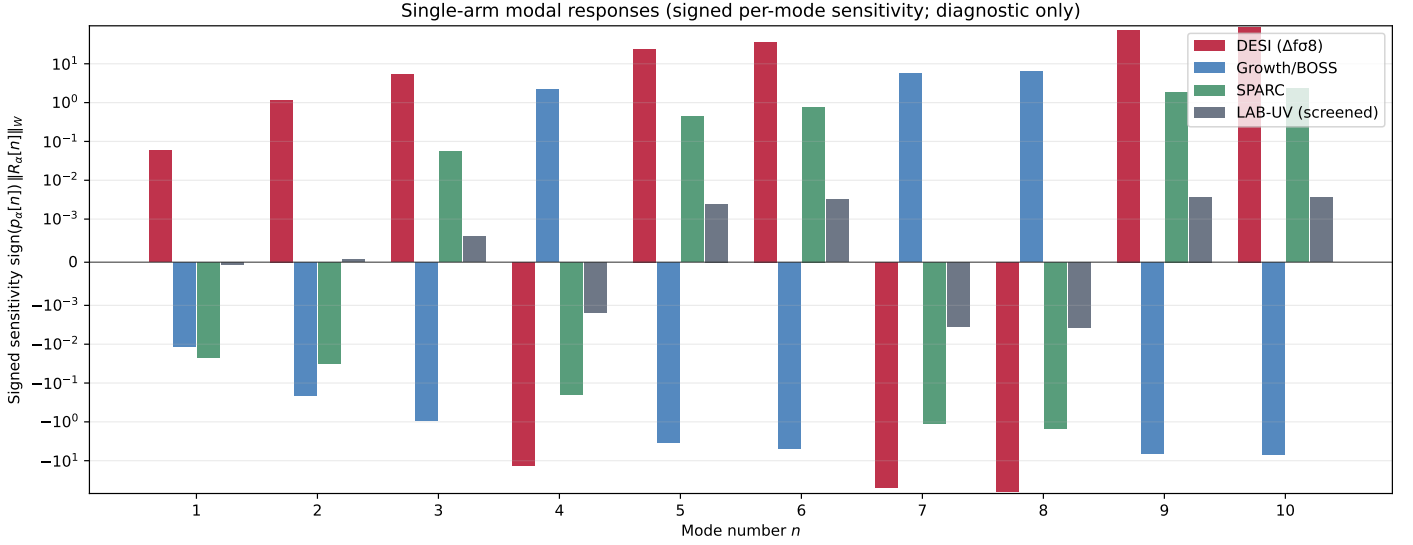


Figure 8: Signed per-mode sensitivities for the separate response arms. The UV-screened laboratory arm remains strongly suppressed.

scalar boundary branches are supplied. Moving endpoints, brane bending, or localized kinetic terms would add boundary contributions and require a new derivation.

The Anti-Zeno response table can likewise be composed with the protocol-time dictionary, but it remains an interface hypothesis until its action, units, and independent calibration are specified.

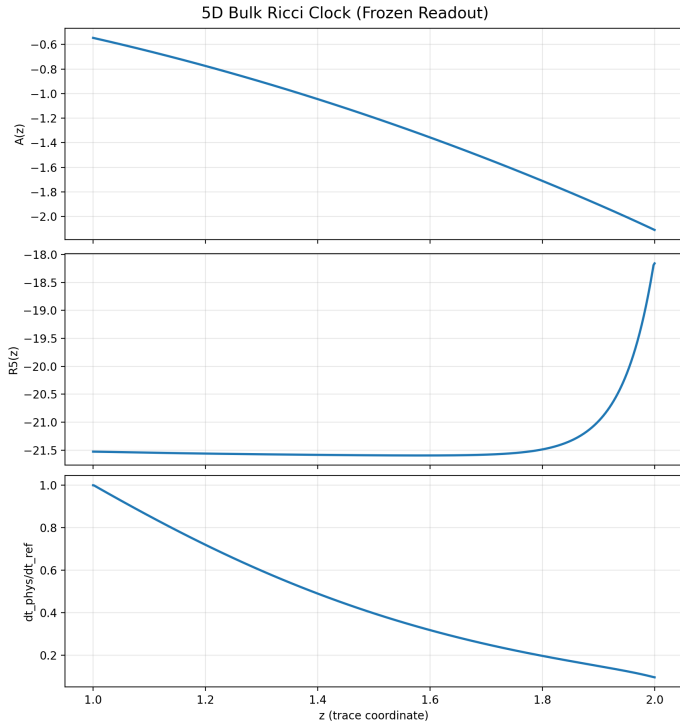


Figure 9: Legacy warp, curvature, and protocol artefact, retained for provenance. Its curvature used the mislabeled stored derivative and is not the corrected $\widehat{R}_5$ of Eq. (61).

13 Discussion and validation path

The effective reconstruction establishes a nontrivial positive result: the actual frozen geometry admits a ghost-free Einstein–scalar completion on the certified finite interval. It also explains why the deformation passed unchanged into downstream arms despite the preconditioner label error. What it does not do is make $K(\phi)$ and $V(\phi)$ prior predictions; they were inferred from the achieved profiles.

A second positive result is available under a sharper evidence label. The compact-interval probe completion derives a normalized matter coupling without using any observational residual or historical fitted coefficient. Its β_0 and mode tower are consequences of the declared interval and Neumann benchmark, not unique consequences of the bulk geometry alone.

Several previously proposed steps are now executable rather than aspirational. The minimal positive Robin boundary-potential family has been mapped; the electromagnetic radial gauge has been corrected; the lapse constraint supplies a scalar–photon vertex; the bulk photon supplies an independently checked spectral comb; and the Wilson analysis fails closed at the missing link ensemble. The scalar and photon combs share one physical length and can therefore be falsified jointly once that scale and a boundary action are fixed. None of those calculations selects those missing inputs. The remaining forward programme is therefore:

1. compress the tabulated $K(\phi)$ and $V(\phi)$ into a compact analytic form without using observational arms;
2. freeze that action, boundary conditions, and solver tolerances;

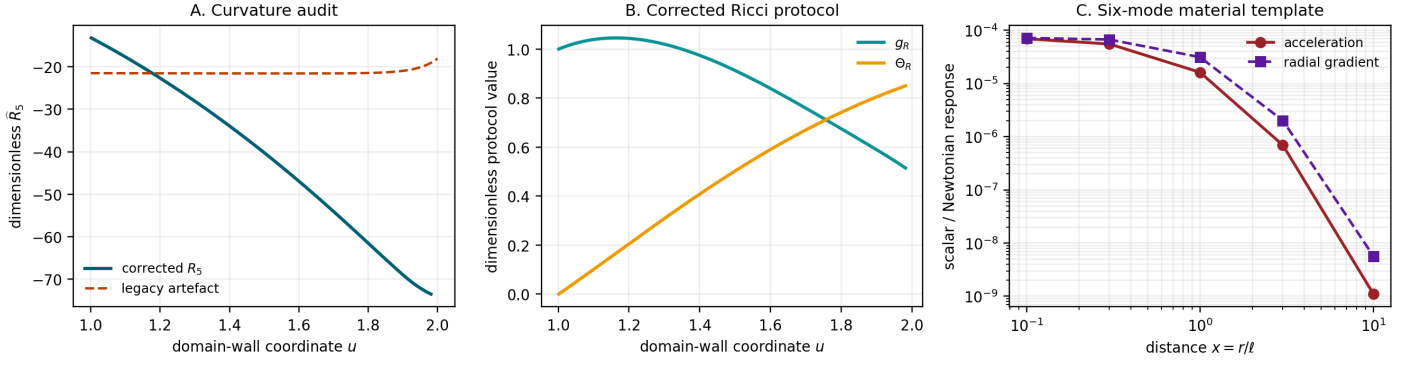


Figure 10: Corrected Ricci/material bridge. Left: corrected curvature from the certified effective action versus the legacy series. Centre: dimensionless curvature cadence and accumulated protocol coordinate. Right: acceleration and tidal templates of the six positive Neumann modes; the physical horizontal scale remains r/ℓ .

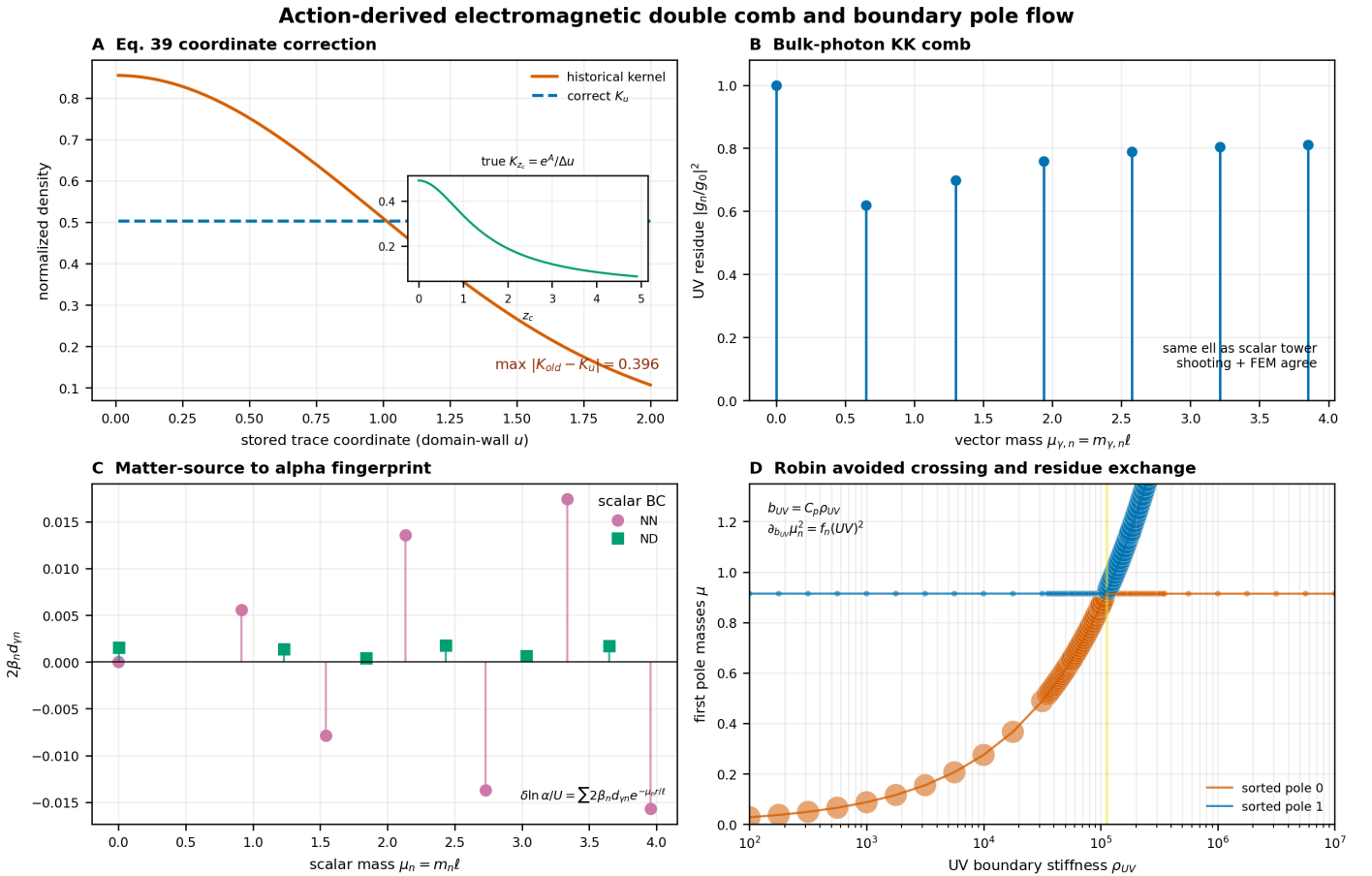


Figure 11: Coordinate-correct electromagnetic completion and double-comb fingerprint. (A) The historical coordinate-mixed profile against the flat bulk-photon density in domain-wall gauge, with the true conformal density inset. (B) Photon Kaluza–Klein masses and UV charge residues. (C) Action-derived source-to- α coefficients for the NN and ND scalar branches. (D) Robin pole motion and UV-residue exchange across the avoided crossing. All axes are dimensionless; no physical boundary coefficient, length scale, source, atomic sensitivity, or detection is inferred.

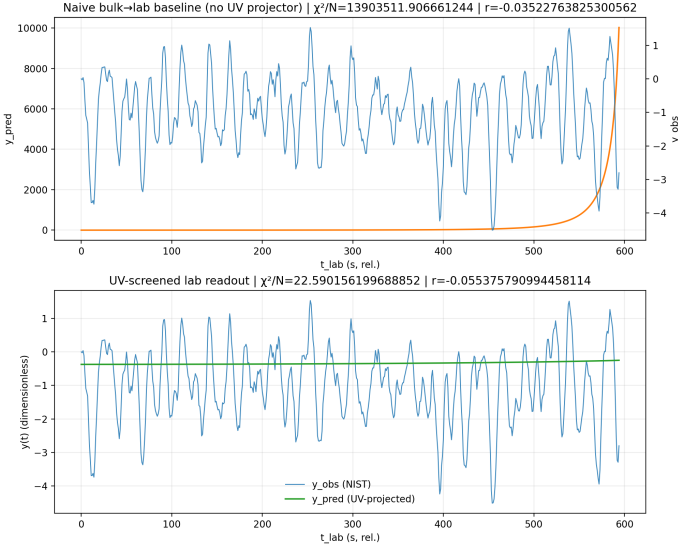


Figure 12: Historical coordinate-mixed clock projections retained for provenance. Their poor correlation is not a detection and is not a test of the corrected bulk-Maxwell response.

3. regenerate A , ϕ , and the canonical χ without reading the old trace;
4. derive the boundary/junction action that fixes a point in the mapped Robin family, and fix ℓ from independent physics;
5. export thermalized SU(3) link ensembles so that the implemented rectangular-loop, static-potential, Creutz, and continuum pipeline can run;
6. recompute the gauge-invariant spectrum with a confining infrared completion and frozen boundary conditions;
7. compare the frozen matter force with fifth-force and post-Newtonian constraints without adding post-hoc screening;
8. drive the derived material transfer law with a calibrated source and test it against a mechanically independent finite-element model and a held-out source distance;
9. derive a four-dimensional cosmological interface before promoting the current radial dictionary to a growth law;
10. choose the bulk- or brane-photon realization prospectively, including photon boundary data, possible localized kinetic terms, source geometry, and atomic response coefficients; and
11. freeze every model and nuisance rule before an independent galaxy, survey-release, or clock-session holdout.

Passing this programme would promote the current operational unification into a forward predictive model.

14 Conclusions

The original work achieved a reproducible shared-background instrument: one frozen warp geometry and deformation feed hadronic, galactic, cosmological, laboratory, and response-operator calculations. The revision identifies that achievement explicitly rather than discarding it.

The previously declared canonical polynomial action and derivative labels do not generate the frozen trace as stated. A geometry-preserving inverse reconstruction supplies the missing bulk completion. Its kinetic function is positive throughout the certified interval, its canonical representation closes the full Einstein–scalar equations at floating-point precision, and it recovers the operational deformation with correlation 0.999999905.

An explicitly conditional compact-interval reduction additionally produces a matter interaction rather than merely naming one: it gives $\beta_n = \sqrt{I_g/3} f_n$, with $\beta_0 = 0.0542901$ and conditional unscreened strength $2\beta_0^2 = 5.89483 \times 10^{-3}$. This result was obtained blind to the observational arms and independently checked numerically. Its physical realization still depends on the boundary/junction completion and on the scale ℓ ; it is neither a detection nor a unique prediction of the bulk trace. Its Neumann massless branch is ruled out in the unscreened universal reading; the retained positive-mode ratios and material transfer equation are therefore a benchmark for a prospectively rederived boundary completion, not permission to delete only the failed zero mode after comparison. The endpoint audit sharpens that warning: the simplest IR-Dirichlet change leaves a strongly coupled ultralight mode, while UV-Dirichlet branches make the declared point probe vanish. The minimal positive Robin phase map adds a stronger result: positive IR stiffness alone has an ultralight ceiling, while UV stiffness moves both poles and residues through an avoided crossing according to the verified Hellmann–Feynman identity. A boundary action, rather than observational performance, must choose the physical realization.

The Ricci and Wilson audits close two further provenance gaps. The corrected $\hat{R}_5$ supplies a dimensionless curvature cadence but not laboratory seconds, and the old Wilson-labelled number is an Einstein-frame endpoint conversion with external α' , not a world-sheet or lattice area law. Using a QCD scale as a compactification length would require a new UV matching relation and cannot be inferred by chaining the present files.

The electromagnetic audit supplies a constructive bridge and one necessary correction. The functional conformal density $K_{z_c} \propto e^A Z(\chi) |f_\gamma|^2$ follows from a bulk Maxwell action, but the historical numerical profile mixed the domain-wall and conformal coordinates. The corrected reduction gives an invariant measure, the action-derived coefficients $d_{\gamma n}$, and a bulk-photon Kaluza–Klein comb. Its shared ℓ with the scalar tower defines a correlated double-comb target rather than two freely aligned spectra. A brane-localized photon gives a different classical result, and neither branch has yet been selected.

The scalar ratio, SPARC performance, BOSS comparison, DESI marginal diagnostic, and UV clock projection remain reproducible outputs. They are now labelled according to what the current evidence establishes. The new retrospective SPARC split is better than a weak Newtonian comparator but worse than the one-parameter RAR comparator; DESI gives no statistical preference; and the Wilson route correctly reports missing input rather than manufacturing a string tension. This sharper separation converts an overextended single-action claim into an executable, falsifiable research programme while preserving the substantial computational structure already built.

Data and code availability

The machine-readable verification bundle is available at https://github.com/RAPIDENN/HOLO_runner and archived under

[doi:10.5281/zenodo.18224589](https://doi.org/10.5281/zenodo.18224589). The effective reconstruction certificate, tabulated action, sealed audit, and exact control flow are included under `first_principles_audit/`. The conditional interaction, its zero-input declaration, finite-element spectrum, independent shooting comparison, Robin phase map, coordinate-correct electromagnetic completion, double-comb fingerprint, Wilson input contract, and present-data diagnostics are stored there as machine-readable artefacts. Third-party SPARC, BOSS, Planck, DESI, and NIST data remain subject to their original distribution policies. The repository bundle records derived metrics and provenance rather than redistributing restricted measurements.

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